TWO MORPHISMS OF SCHEMES WITH 2-ISOMORPHIC SMALL MORPHISMS OF RINGED TOPOI ARE EQUAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument, including the author's opening remark that careful argument is needed; the Xy-pic diagram is delivered as written.

Editorial note. The label lemma-faithful occurs in no other staged Stacks chapter, so no cross-chapter restatement had to be weighed for this card.

Editorial note. The single labelled reference occurring inside the counted spans points to a chapter of the Project that is not staged in this card's source tree; it is carried in assembly.external_references as a non-load-bearing navigation pointer recorded at the invoking line.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Small morphism of ringed topoi attached to a morphism of schemes. Same chapter, the morphism (a_{small}, a^sharp) : (X_etale, O_X) -> (Y_etale, O_Y) associated to a morphism a : X -> Y. Supplies the objects compared by the assumed 2-isomorphism.
- Subsheaves of the final sheaf on a site and the sheaf h_V represented by an open subset. Same chapter, the sheaf h_V on Y_etale for an open V of Y, a subsheaf of the final sheaf, and the identity a_{small}^{-1}h_V = h_{a^{-1}(V)}. Supplies the argument by which the 2-isomorphism is forced to be the identity on the underlying topological spaces.
- Sections of the structure sheaf as maps h_V -> O_Y, and a^sharp as the structural comparison map. Same chapter, the description of a section f in O_Y(V) by the induced map of sheaves h_V -> O_Y and the commutativity of the square defining a 2-isomorphism of ringed topoi. Supplies the diagram whose commutativity gives a_{small}^sharp(f) = b_{small}^sharp(f).

Source reference. Original section 04IB: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 5336–5336. Cross-chapter navigation pointer invoked at proof lines 5335-5336, where the author refers to the definition of a 2-isomorphism of ringed topoi. The target chapter is not staged in this card's source tree, so the pointer records the invoking line; it only resolves the reference for the reader, and no bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let X, Y be schemes. Any two morphisms $a, b : X \to Y$ of schemes for which there exists a 2-isomorphism $(a_{small}, a^\sharp_{small}) \cong (b_{small}, b^\sharp_{small})$ in the 2-category of ringed topoi are equal.*

Proof. Let us argue this carefully since it is a bit confusing. Let $t : a^{-1}_{small} \to b^{-1}_{small}$ be the 2-isomorphism. Consider any open $V \subset Y$. Note that h_V is a subsheaf of the final sheaf $*$. Thus both $a^{-1}_{small}h_V = h_{a^{-1}(V)}$ and $b^{-1}_{small}h_V = h_{b^{-1}(V)}$ are subsheaves of the final sheaf. Thus the isomorphism

$$t : a^{-1}_{small}h_V = h_{a^{-1}(V)} \to b^{-1}_{small}h_V = h_{b^{-1}(V)}$$

has to be the identity, and $a^{-1}(V) = b^{-1}(V)$. It follows that a and b are equal on underlying topological spaces. Next, take a section $f \in \mathcal{O}_Y(V)$. This determines and is determined by a map of sheaves of sets $f : h_V \to \mathcal{O}_Y$. Pull this back and apply t to get a commutative diagram

$$\begin{array}{ccccc}
h_{b^{-1}(V)} & = & b^{-1}_{small}h_V & \xleftarrow{t} & a^{-1}_{small}h_V & = & h_{a^{-1}(V)} \\
 & & \downarrow{\scriptstyle b^{-1}_{small}(f)} & & \downarrow{\scriptstyle a^{-1}_{small}(f)} & & \\
 & & b^{-1}_{small}\mathcal{O}_Y & \xleftarrow{t} & a^{-1}_{small}\mathcal{O}_Y & & \\
 & & & {\scriptstyle b^\sharp}\searrow \quad \swarrow{\scriptstyle a^\sharp} & & & \\
 & & & \mathcal{O}_X & & &
\end{array}$$

where the triangle is commutative by definition of a 2-isomorphism in Modules on Sites, Section 04IB. Above we have seen that the composition of the top horizontal arrows comes from the identity $a^{-1}(V) = b^{-1}(V)$. Thus the commutativity of the diagram tells us that $a^\sharp_{small}(f) = b^\sharp_{small}(f)$ in $\mathcal{O}_X(a^{-1}(V)) = \mathcal{O}_X(b^{-1}(V))$. Since this holds for every open V and every $f \in \mathcal{O}_Y(V)$ we conclude that $a = b$ as morphisms of schemes. □